Calculus I

Lecture #3

Trigonometric Functions:

In this lecture we study the trig. functions:

1. $y = \sin x$ 2. $y = \cos x$
3. $y = \tan x = \frac{\sin x}{\cos x}$ 4. $y = \cot x = \frac{\cos x}{\sin x}$
5. $y = \sec x = \frac{1}{\cos x}$ 6. $y = \csc x = \frac{1}{\sin x}$

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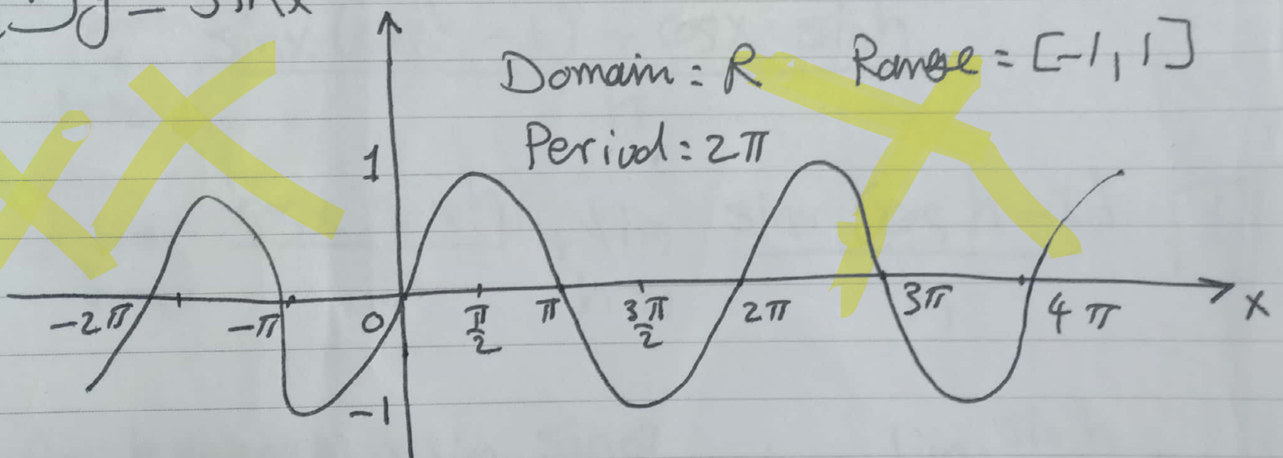
1. $y = \sin x$



Notice that:

* $\sin\theta = 0 \Longrightarrow \theta = 0, \pm\pi, \pm 2\pi, \dots \Longrightarrow \theta = n\pi, n \in \mathbb{Z}$
* $\lim_{\theta \to \infty} \sin\theta$ does not exist (Why?)

Rule:

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\sin(f(x)) = f'(x)\cos(f(x))$$

Proof:

$$y(x) = \sin x$$

$$y'(x) = \lim_{h\to 0}\frac{y(x+h)-y(x)}{h} = \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h}$$

As known $\quad \sin(A\pm B) = \sin A\cos A \pm \cos A\sin B$

then $\quad y'(x) = \lim_{h\to 0}\dfrac{\sin x\cosh + \cos x\sin h - \sin x}{h}$

$$= \lim_{h\to 0}\frac{\sin x(\cos h-1)+\cos x\sin h}{h}$$

$$= \lim_{h\to 0}\frac{\cos x\sin h}{h} + \lim_{h\to 0}\frac{\sin x(\cos h-1)}{h} \quad \boxed{1}$$

As known $\Rightarrow \lim_{\theta\to 0}\dfrac{\sin\theta}{\theta} = 1 \Rightarrow \lim_{h\to 0}\dfrac{\sin h}{h} = 1$

$$\cos h - 1 = -2\sin^2 h/2$$

$$\Rightarrow \frac{\cos h-1}{h} = -\frac{2\sin^2 h/2}{h} = -\sin h/2\ \frac{\sin h/2}{h/2}$$

Back to $\boxed{1}$

$$y'(x) = \lim_{h\to 0} \cos x \left(\frac{\sin h}{h}\right) + \lim_{h\to 0} \sin x \left(-\sin h/2 \; \frac{\sin h/2}{h/2}\right)$$

$$= \cos x \lim_{h\to 0}\left(\frac{\sin h}{h}\right) + \sin x \lim_{h\to 0}\left(-\sin \frac{h}{2}\right)\left(\frac{\sin h/2}{h/2}\right)$$

$$= \cos x \quad (1) + \sin x \quad (0)(1)$$

$$= \cos x + 0 = \cos x$$

$$\Longrightarrow \frac{d}{dx} \sin x = \cos x$$

Similarly, you can show that

$$\Longrightarrow \cancel{\frac{d}{dx} \cos x}$$

$$\frac{d}{dx} \sin(f(x)) = f'(x) \cos(f(x))$$

Example: Find $y'(x)$

ⓐ $y = \sin(5x^3 - 10x + 7)$ ⓑ $y = \sqrt{\sin e^{4x}}$

Solution:

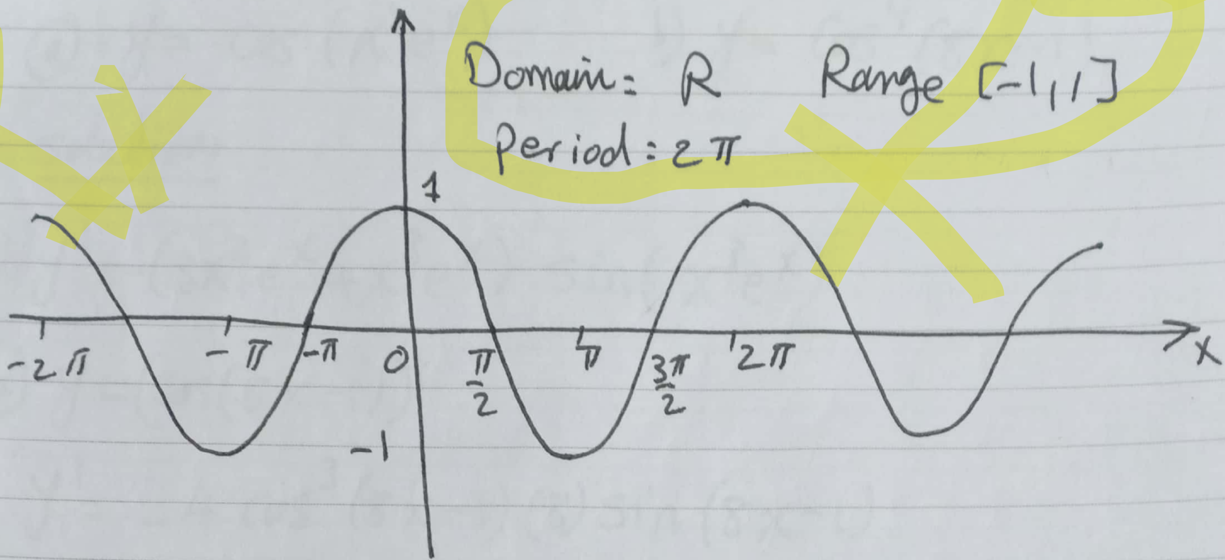
ⓐ $y' = (15x^2 - 10)\cos(5x^3 - 10x + 7)$

ⓑ $y' = \frac{1}{2}(\sin e^{4x})^{-\frac{1}{2}}(4e^{4x} \cos e^{4x})$

2 $y = \cos x$

Domain: $\mathbb{R}$ Range $[-1, 1]$

Period: 2π



Notice that:

* $\cos\theta = 0 \Rightarrow \theta = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \dots \quad \theta = (2n+1)\frac{\pi}{2}$
* $\lim\limits_{x\to\infty} \cos x$ does not exist.

Rule:

$$\frac{d}{dx}\cos x = -\sin x \qquad \frac{d}{dx}\cos(f(x)) = -f'(x)\sin(f(x))$$

Proof. $\Longrightarrow$ Exercise

Remark: $y = \sin\left(\frac{x}{3}\right) \Longrightarrow \text{period} = 2\pi(3) = 6\pi$

$y = \cos(3x) \Rightarrow \text{period} = \frac{2\pi}{3}$

Example: Find $y'(x)$

ⓐ $y = \cos(x^3 e^x)$ ⓑ $y = \cos^4(8x-1)$

solution:

ⓐ $y' = -(3x^2 e^x + x^3 e^x)\sin(x^3 e^x)$

ⓑ $y = (\cos(8x-1))^4$

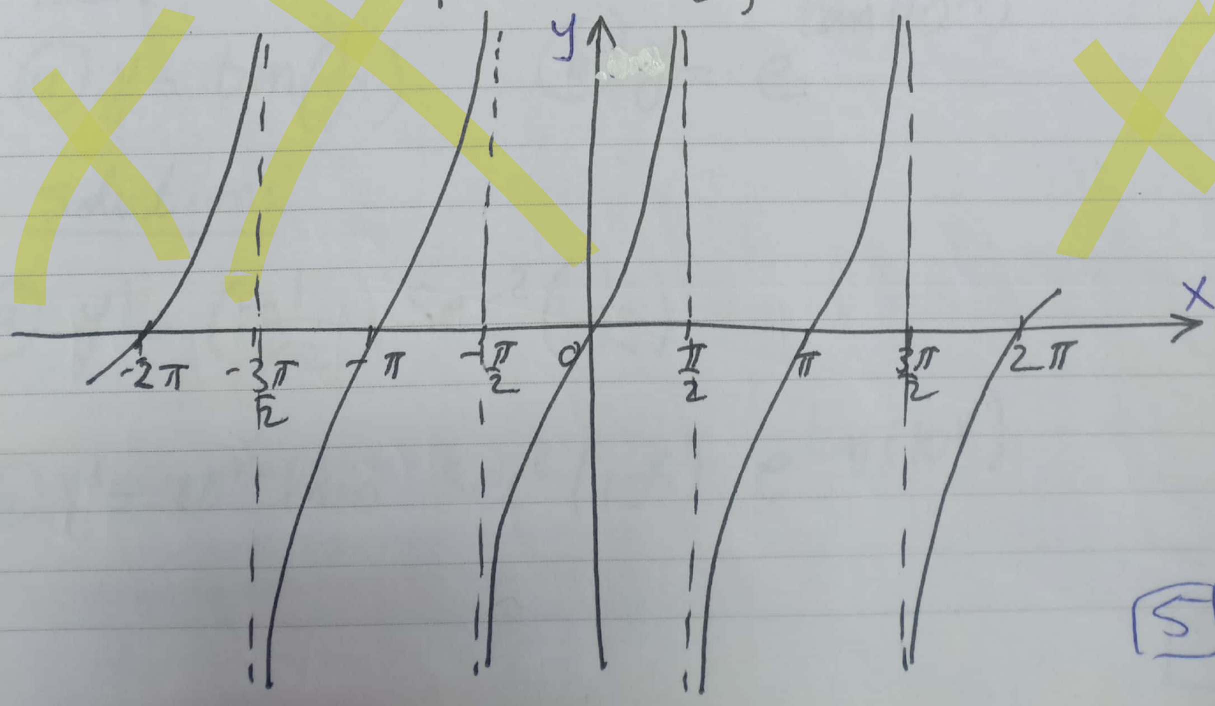
$y' = -4\cos^3(8x-1)(8)\sin(8x-1)$

$= -32\sin(8x-1)\cos^3(8x-1)$

$\boxed{3}$ $y = \tan x = \dfrac{\sin x}{\cos x}$

Domain $= R - \{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots\}$

$\Rightarrow R - \{\pm(2n+1)\frac{\pi}{2},\ n = 0, 1, 2, 3, \ldots\}$



Range: $]-\infty, \infty[$

Period: π

Rule:

$$\frac{d}{dx}\tan x = \sec^2 x \qquad \frac{d}{dx}\tan f(x) = f'(x)\sec^2(f(x))$$

proof: $y = \tan x = \dfrac{\sin x}{\cos x}$

$$y' = \frac{\cos x \cos x + \sin x \sin x}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

Example: Find $y'(x)$

ⓐ $y = \tan\left(\frac{1}{x}\right)$ ⓑ $y = e^{\tan(10^x)}$

Solution:

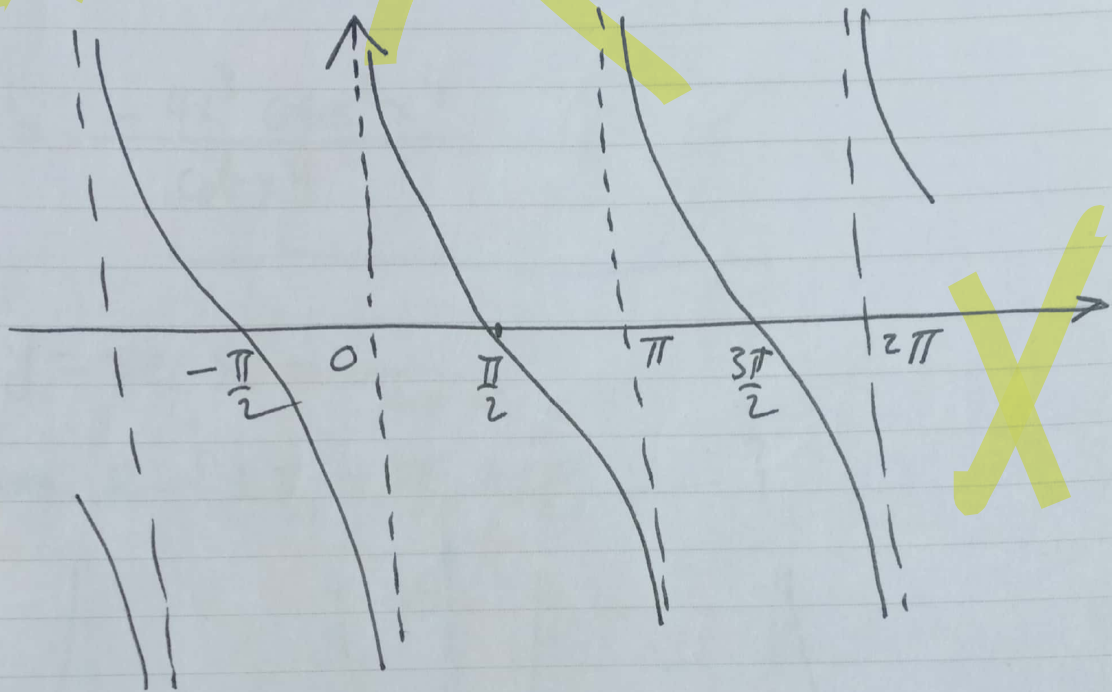
ⓐ $y' = \left(-\frac{1}{x^2}\right)\sec^2\left(\frac{1}{x}\right)$

ⓑ $y' = 10^x \ln 10 \, \sec^2(10^x)\, e^{\tan(10^x)}$

[4] $y = \cot x = \dfrac{\cos x}{\sin x}$

Domain: $R - \{0, \pm\pi, \pm 2\pi, ---\}$

$\Rightarrow R - \{\pm n\pi, n = 0, 1, 2, 3, --\}$



Range: $]-\infty, \infty[$

Period: π

$\dfrac{d}{dx}\cot x = -\operatorname{cosec}^2 x \qquad \dfrac{d}{dx}\cot f(x) = -f'(x)\operatorname{cosec}^2(f(x))$

Proof: Exercise

Example: Find $y'(x)$

ⓐ $y = \cot 4^x$ ⓑ $y = \ln(\cot x^4)$

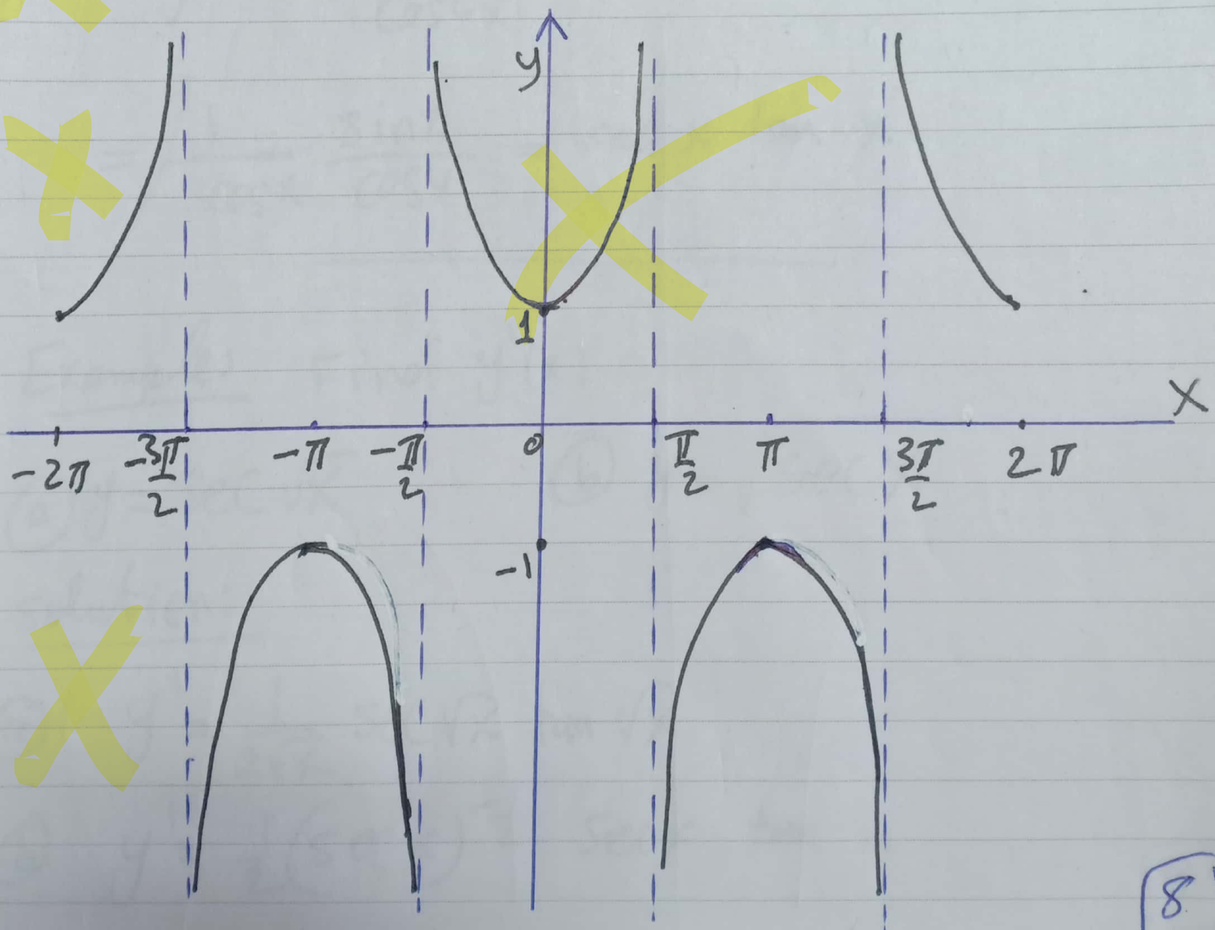
Solution:

ⓐ $y' = -4^x \ln 4 \ \operatorname{cosec}^2 4^x$

ⓑ $y' = \dfrac{-4x^3 \operatorname{cosec}^2 x^4}{\cot x^4}$

$\boxed{5}$ $y = \sec x = \dfrac{1}{\cos x}$

Domain: $R - \{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, ----\}$



Range: $]-\infty,-1] \cup [1,\infty[$

Period: 2π

$$\frac{d}{dx}\sec x = \sec x \tan x$$

$$\frac{d}{dx}\sec(f(x)) = f'(x)\,\sec(f(x))\,\tan(f(x))$$

Proof: $y = \sec x = \dfrac{1}{\cos x}$

$$y' = \frac{0 + \sin x}{\cos^2 x}$$

$$= \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x$$

Examples: Find $y'(x)$

ⓐ $y = \sec\sqrt{x}$ ⓑ $y = \sqrt{\sec x}$

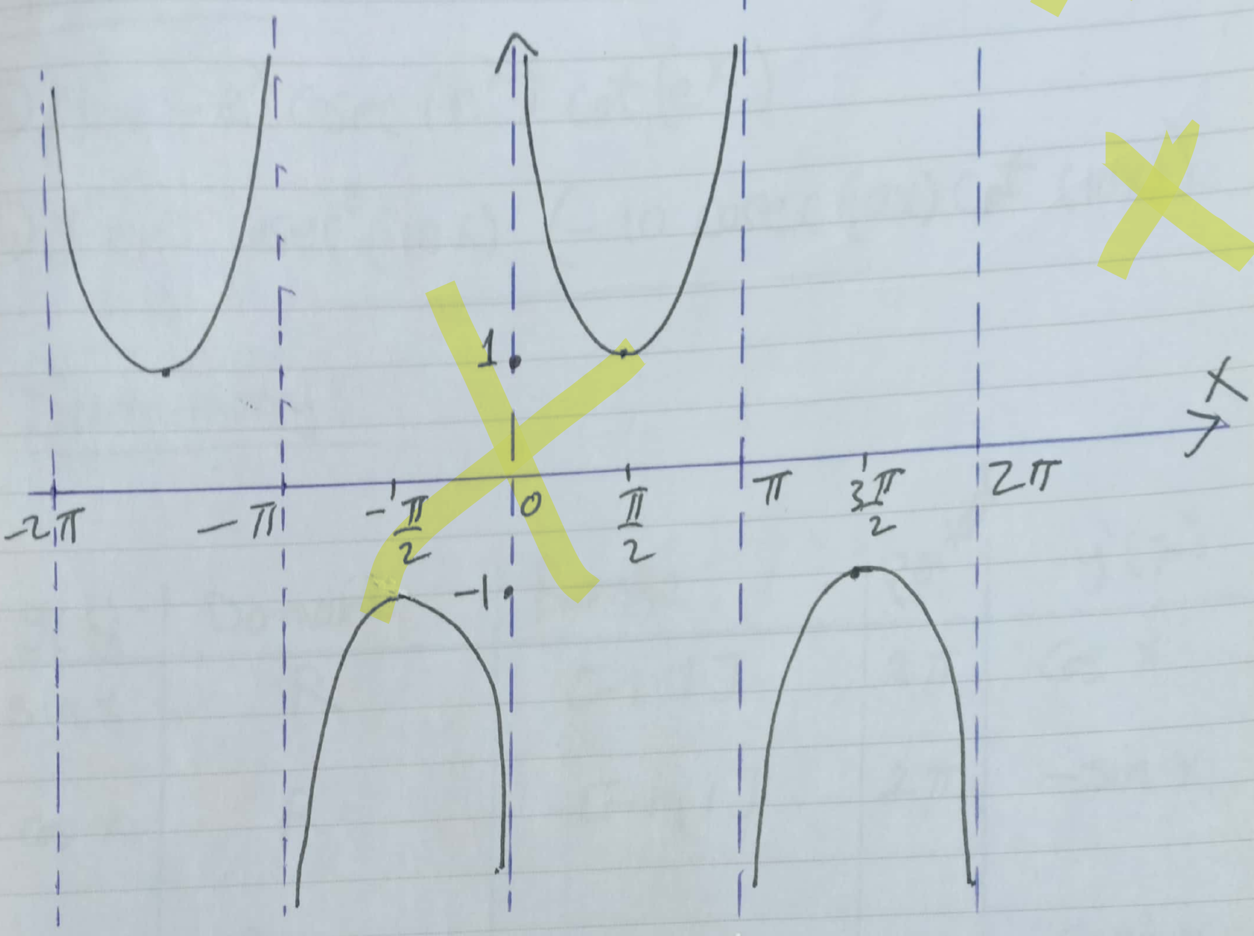
solution:

ⓐ $y' = \dfrac{1}{2\sqrt{x}} \sec\sqrt{x} \tan\sqrt{x}$

ⓑ $y' = \frac{1}{2}(\sec x)^{-\frac{1}{2}} \sec x \tan x$

⑥ $y = \operatorname{cosec} x = \dfrac{1}{\sin x}$

Domain: $\mathbb{R} - \{0, \pm\pi, \pm 2\pi, ----\}$



Range: $]-\infty, -1] \cup [1, \infty[$

Period: 2π

$\dfrac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$

$\dfrac{d}{dx} \operatorname{cosec}(f(x)) = -f'(x) \operatorname{cosec}(f(x)) \cot(f(x))$

proof $\Rightarrow$ Exercise

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Example: Find $y'(x)$

ⓐ $y = \operatorname{cosec}(e^x)$ ⓑ $y = \operatorname{cosec}^3(10x)$

Solution:

ⓐ $y' = -e^x \operatorname{cosec}(e^x)\cot(e^x)$

ⓑ $y' = 3\operatorname{cosec}^2(10x)\left(-10\operatorname{cosec}(10x)\cot(10x)\right)$

Summary:

$y(x)$	Domain	Range	period	$y'(x)$
$\sin x$	R	$[-1,1]$	2π	$\cos x$
$\cos x$	R	$[-1,1]$	2π	$-\sin x$
$\tan x$	$R-\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \dots\}$	R	π	$\sec^2 x$
$\cot x$	$R-\{0, \pm\pi, \pm 2\pi, \dots\}$	R	π	$-\operatorname{cosec}^2 x$
$\sec x$	$R-\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \dots\}$	$]-\infty,-1] \cup [1,\infty[$	2π	$\sec x \tan x$
$\operatorname{cosec} x$	$R-\{0, \pm\pi, \pm 2\pi, \dots\}$	$]-\infty,-1] \cup [1,\infty[$	2π	$-\operatorname{cosec} x \cot x$

Example: Find the period of

ⓐ $y = \sin \frac{x}{3}$ ⓑ $y = \cos 2x$ ⓒ $y = \tan \frac{x}{4}$

ⓓ $y = \csc 3x$ ⓔ $y = \cot 10x$

Solution:

ⓐ $p = 3(2\pi) = 6\pi$

ⓑ $p = \frac{2\pi}{2} = \pi$

ⓒ $p = 4(\pi) = 4\pi$

ⓓ $p = \frac{2\pi}{3}$

ⓔ $p = \frac{\pi}{10}$

Example: Find $y'(x)$

ⓐ $y = \cos^3\left(\frac{1-x}{1+x}\right)$ ⓑ $y = \sin(x^2 e^{-x})$

ⓒ $y = \sqrt[3]{\tan(\pi^x)}$ ⓓ $y = e^{\cot \sqrt{x}}$

ⓔ $y = \ln(\sec(10x-1))$ ⓕ $y = \sqrt{\csc x^\pi}$

Solution:

ⓐ $y = \cos^3\left(\frac{1-x}{1+x}\right)$

$y' = 3\cos^2\left(\frac{1-x}{1+x}\right)\left(-\sin\left(\frac{1-x}{1+x}\right)\right)\left[\frac{-(1+x)-(1-x)}{(1+x)^2}\right]$

ⓑ $y = \sin(x^2 e^{-x})$

$y' = \cos(x^2 e^{-x})\ [2xe^{-x} - x^2 e^{-x}]$

ⓒ $y = (\tan(\pi^x))^{1/3}$

$y' = \frac{1}{3}(\tan(\pi^x))^{-2/3}(\sec^2 \pi^x)(\pi^x \ln \pi)$

ⓓ $y = e^{\cot\sqrt{x}}$

$y' = e^{\cot\sqrt{x}}\left(-\operatorname{cosec}^2\sqrt{x}\right)\left(\frac{1}{2\sqrt{x}}\right)$

ⓔ $y' = \frac{10\sec(10x-1)\cdot\tan(10x-1)}{\sec(10x-1)} = 10\tan(10x-1)$

ⓕ $y' = \frac{1}{2}(\operatorname{cosec} x^\pi)^{-\frac{1}{2}}(-\operatorname{cosec} x^\pi \cot x^\pi)(\pi x^{\pi-1})$

$\boxed{13}$